The Defense Trilemma: Why Prompt Injection Defense Wrappers Fail?

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Abstract

We prove that no continuous, utility-preserving wrapper defense—a function $D\colon X \to X$ that preprocesses inputs before the model sees them—can make all outputs strictly safe for a language model with connected prompt space, and we characterize exactly where every such defense must fail. We establish three results under successively stronger hypotheses: *boundary fixation*—the defense must leave some threshold-level inputs unchanged; an *ε-robust constraint*—under Lipschitz regularity, a positive-measure band around fixed boundary points remains near-threshold; and a *persistent unsafe region*—under a transversality condition, a positive-measure subset of inputs remains strictly unsafe. These constitute a *defense trilemma*: continuity, utility preservation, and completeness cannot coexist. We prove parallel discrete results requiring no topology, and extend to multi-turn interactions, stochastic defenses. The results do not preclude training-time alignment, architectural changes, or defenses that sacrifice utility. The full theory is mechanically verified in Lean 4 with Mathlib `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs` (46 files, $\sim$360 theorems, no admitted proofs, three standard axioms) and illustrated empirically on three LLMs [17].

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities? Most current defense work implicitly assumes yes. Input classifiers that flag suspicious prompts [1], constitutional rewriting pipelines [3], and input sanitization layers [7] all share the same structure: a function $D\colon X \to X$ that preprocesses prompts before the model sees them, mapping unsafe inputs to safe equivalents while leaving safe inputs unchanged. We prove the answer is no, under two constraints. If the defense is *continuous* (similar prompts produce similar rewrites) and *utility-preserving* (safe prompts pass through unchanged), it cannot be *complete* (make every output safe). These three properties form a **defense trilemma**: any two can coexist, but not all three (Figure 2).

The impossibility is not about specific attacks or clever prompt engineering. It arises from the *geometry* of the prompt space itself: in a connected space, the safe region is open but not closed, so any continuous defense that fixes safe inputs must also fix points on the safety boundary. Under

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successively stronger hypotheses, we establish three results with progressively stronger conclusions (Figure 3):

Boundary fixation (Theorem 4.1). The defense must fix at least one boundary point, i.e., a prompt where alignment deviation equals the threshold exactly—passing it through with no remediation.

ε-robust constraint (Theorem 5.1). Under Lipschitz regularity, the defense cannot uniformly reduce alignment deviation far below τ near the fixed boundary point. For any x within distance δ of the fixed point z:

$$f(D(x)) \;\geq\; \tau - LK\,\delta. \tag{1}$$

Persistent unsafe region (Theorem 6.3). Under a transversality condition, the alignment surface rises faster than the defense can pull it down, leaving a positive-measure region that remains unsafe:

$$f(D(x)) > \tau \quad \text{for all } x \in \mathcal{S}, \qquad \mu(\mathcal{S}) > 0. \tag{2}$$

From discrete to continuous. All three results apply to continuous interpolants of discrete data. The Tietze extension theorem guarantees that any finite set of behavioral observations in a normal space admits continuous extensions, and the impossibilities hold for every such extension (Theorem 8.1).

Scope and limitations. Our results apply specifically to *continuous, utility-preserving wrapper defenses* on *connected* prompt spaces. They do *not* preclude effective safety through other mechanisms, including:

- training-time alignment (RLHF, DPO, constitutional AI training),
- architectural changes to the model itself,
- discontinuous defenses (e.g., hard blocklists or discrete classifiers),
- output-side filters, ensemble defenses, or human-in-the-loop review,
- adaptive-threshold systems,
- multi-component systems whose classifiers may reject or redirect inputs rather than preserving utility on every prompt.

In short, our theorems cover only a single continuous wrapper $D\colon X \to X$ that preprocesses inputs; any mechanism outside this class is not constrained by our results.

All three conditions—continuity, utility preservation, and connectedness—are individually necessary; we give counterexamples for each (Section C).

Contributions.

1. **Boundary fixation** (Theorem 4.1): any continuous, utility-preserving defense on a connected Hausdorff space must fix a point z with $f(z) = \tau$. Relaxed to score-preserving and ε-approximate variants (Theorems 4.4 and 4.5).
2. **ε-robust constraint** (Theorem 5.1): under Lipschitz regularity, $f(D(x)) \geq \tau - LK\,\mathrm{dist}(x, z)$ for all x. A positive-measure band near z is constrained (Theorem 5.2).
3. **Persistent unsafe region** (Theorem 6.3): under a transversality condition ($G > \ell(K+1)$, where ℓ is the defense-path Lipschitz constant), a positive-measure set remains strictly above τ after defense.
4. **Quantitative bounds**: explicit volume lower bound (Theorem 7.1), cone measure bound (Theorem 7.2), and an asymmetric defense dilemma (Theorem 7.3).
5. **Discrete defense dilemma** (Theorem 8.3): on finite sets, completeness forces non-injectivity (information loss); injectivity forces incompleteness.
6. **Extensions**: multi-turn (Theorem 9.1), stochastic (Theorem 9.2), and pipeline (Theorem 9.3) settings.
7. **Lean 4 formalization**: 46 files, $\sim$360 theorems, zero `sorry` statements, three standard axioms. Full proofs in Section B; artifact in Section I.

Table 1: Positioning among impossibility results (top) and empirical findings independently consistent with our predictions (bottom).

Result	**Target**	**Method**	**Impossibility**	**Verified**
Tsipras et al. [21]	Classifier	PAC / Gaussian	Robustness + accuracy	No
Fawzi et al. [11]	Classifier	Lipschitz geom.	Adv. invulnerability	No
Wolpert & Macready [22]	Optimizer	Counting	Universal dominance	No
Cohen et al. [5]	Smoothed clf.	Rand. smoothing	(Certifies bound)	No
This work	**Def. wrapper**	**Topology**	**Cont. + util. + compl.**	**Lean 4**
Empirical findings consistent with our derived predictions				
Our prediction	**Theorem**	**Confirming evidence**		**Pred.**
Diminishing safety returns	Int. Stability	MART [24]; Sleeper Agents [25]		§11
Long context $\to$ exp. harder defense	Cost Asymm.	Many-Shot Jailbreaking [26]; Kim et al. [27]		§11
Agentic tool-use degrades exp.	Pipeline Lip.	InjecAgent [28]; ASB [29]		§11
Temperature boundary instab.	Stoch. Imposs.	Instability of Safety [30]		§11
Quantization preserves deep vulns	Int. Stability	Skoltech [31]; ETH [32]		§11
Merging preserves weaker parent	Int. Stability	Hammoud et al. [33]		§11
Adv. training fragments	Fragment Size	Whack-a-Mole [34]; IRIS [35]		§11
Asymmetric transfer	Transferability	Weak-to-Strong [36]; IRIS [35]		§11
Alignment tax $\propto$ unsafe vol.	Discrete Dil.	Safety Tax [37]; Huang et al. [38]		§11
Patch blast radius	Patching Nonl.	Slingshot [39]; Sleeper Agents [25]		§11

2 Related Work

Adversarial robustness research has established that small perturbations can fool classifiers [20, 6, 9, 14] and that robustness may fundamentally trade off against accuracy [21, 11]. Certified defenses provide guarantees for fixed models [5, 8, 19, 13, 2], and topological perspectives have illuminated the structure of decision boundaries [16]. For LLMs specifically, jailbreaking attacks [23, 4, 10, 12] and automated red-teaming [15, 18] demonstrate persistent vulnerabilities.

Our work differs in kind: rather than studying how failures arise for fixed models or how specific systems can be certified, we impose *universal constraints on the defense map itself*. The closest conceptual precedent is the no-free-lunch framework [22], which proves that no optimizer dominates across all problems. We prove the analogous result for wrapper defenses: under continuity and utility preservation, no defense eliminates all failures.

Table 1 situates our result among comparable impossibility and tradeoff theorems.

3 Formal Framework

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space (a set equipped with a notion of "open" and "closed" sets, such that the whole space and the empty set are both open and closed, arbitrary unions and finite intersections of open sets are open, and a notion of "neighborhoods" of points—that is, points "close to" a given point). An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$, i.e., small changes in the input produce small changes in the score.*

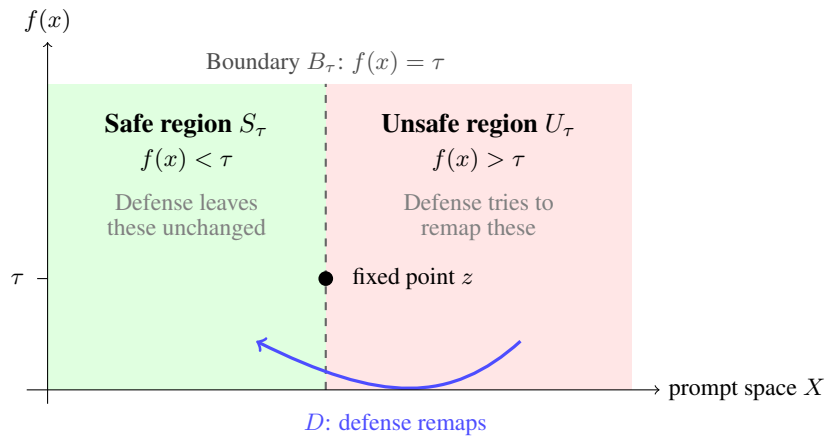

Figure 1: Schematic of the prompt space. The defense D must leave all safe inputs unchanged (utility preservation) and tries to remap unsafe inputs into the safe region. Boundary fixation (Theorem 4.1) proves the defense must also leave some boundary points unchanged, the black dot z passes through with no remediation.

Given a threshold $\tau \in \mathbb{R}$:

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{3}$$
$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{4}$$
$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{5}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$ (safe prompts pass through unchanged), and* complete *if $f(D(x)) < \tau$ for all $x \in X$ (every output is safe).*

The central question is whether a defense can be both utility-preserving and complete. The following sections show that under natural conditions the answer is no.

4 Boundary Fixation

We begin with the most fundamental result. The argument fits in a paragraph: the defense fixes every safe input, so continuity makes its fixed-point set closed. But the safe region S_τ is open (preimage of an open interval under continuous f), and in a connected space a nonempty proper open set is not closed. Hence the fixed-point set cannot stop exactly at the edge of the safe region, it must spill onto the boundary B_τ. Some boundary prompts pass through unchanged.

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$, and this set is nonempty.*

Proof sketch. In a Hausdorff space, $\mathrm{Fix}(D)$ is closed (preimage of the diagonal). By utility preservation, $S_\tau \subseteq \mathrm{Fix}(D)$, so $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. But $S_\tau = f^{-1}((-\infty, \tau))$ is open and not closed (connectedness: a nonempty proper clopen set would disconnect X). Hence $\overline{S_\tau} \supsetneq S_\tau$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (limits of values $<\tau$ cannot exceed τ, and $z \notin S_\tau$ forces $f(z) \geq \tau$) and $D(z) = z$. Full proof in Section B. □

This means that the defense's fixed-point set is too large to avoid the boundary. Utility preservation forces it to contain the safe region; closure forces it to contain the boundary; connectedness ensures the boundary is nonempty. Alternatively: a complete utility-preserving defense would be a continuous retraction $D\colon X \to S_\tau$ (since $D|_{S_\tau} = \mathrm{id}$ and $D(X) \subseteq S_\tau$), but a retract of a Hausdorff

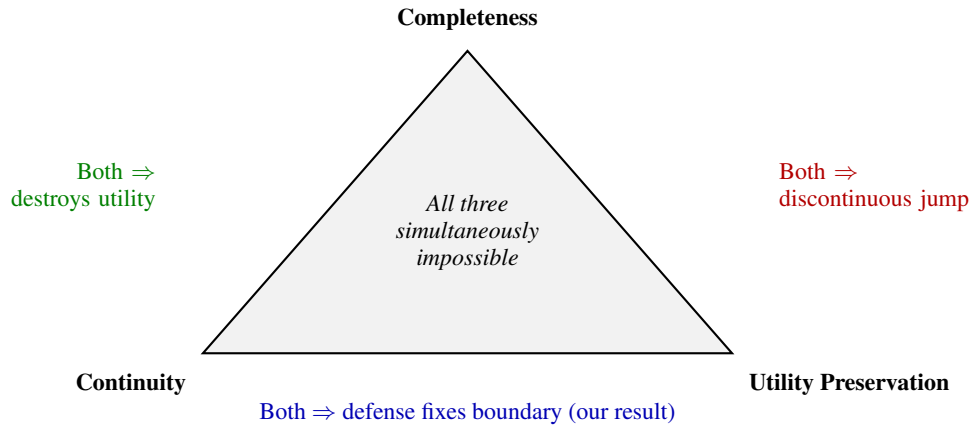

Figure 2: The defense trilemma. Any continuous wrapper defense on a connected space can satisfy at most two of the three properties. The bottom edge is our main result; the other two edges correspond to counterexamples in Section C.

space is closed (the fixed-point set is closed), while S_τ is open and not closed (connectedness)—a contradiction.

Theorem 4.2 (Defense Trilemma)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. No $D\colon X \to X$ can simultaneously be continuous, utility-preserving ($D|_{S_\tau} = \mathrm{id}$), and complete ($f(D(x)) < \tau$ for all x).*

All three hypotheses are individually necessary. A defense can satisfy at most two of the three—the *defense trilemma* (Figure 2). Counterexamples for each dropped hypothesis appear in Section C.

Remark 4.3 (Not all boundary points are fixed)**.** *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points not in $\overline{S_\tau}$ may escape fixation.*

4.1 Relaxed Utility Preservation

Strict utility preservation ($D(x) = x$ for safe x) can be relaxed. The impossibility survives score-preserving rewrites and even approximate score preservation.

Theorem 4.4 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) = \tau$.*

Proof sketch. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{h = 0\}$ is closed and contains $\overline{S_\tau}$, since closed sets contain their boundary points. For $z \in \overline{S_\tau} \setminus S_\tau$, $f(D(z)) = f(z) = \tau$. □

The next result weakens score preservation to approximate:

Theorem 4.5 (ε-Relaxed Utility Preservation)**.** *Under the hypotheses of Theorem 4.4, if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) \geq \tau - \varepsilon$.*

Proof sketch. The set $\{x : h(x) \geq -\varepsilon\}$ is closed and contains S_τ, hence $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = \tau + h(z) \geq \tau - \varepsilon$. □

Remark 4.6 (Why $D(S_\tau) \subseteq S_\tau$ alone is insufficient)**.** *The weakest relaxation—$D(S_\tau) \subseteq S_\tau$ with no score constraint—does allow a complete defense (e.g., a constant map $D(x) = x_0$ to a fixed safe point). But this destroys all semantic content: every prompt produces the same response. The score-preservation conditions formalize the requirement that defense must not destroy utility, without requiring the defense to be the identity.*

5 ε-Robust Constraint

Boundary fixation produces at least one fixed point; Lipschitz regularity makes the failure spread to a neighborhood.

Theorem 5.1 (ε-Robust Defense Constraint)**.** *Under the hypotheses of Theorem 4.1, if (X, d) is a metric space, f is L-Lipschitz, and D is K-Lipschitz, then for the fixed boundary point z:*

$$f(D(x)) \;\geq\; \tau - LK\,\mathrm{dist}(x, z) \qquad \textit{for all } x \in X. \tag{6}$$

Points within distance δ of z remain within $LK\delta$ of threshold.

Proof sketch. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) \leq K\,\mathrm{dist}(x, z)$. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| \leq L \cdot K\,\mathrm{dist}(x, z)$. Full proof in Section B. □

Theorem 5.2 (Positive-Measure ε-Band)**.** *Under the hypotheses of Theorem 5.1, if X is connected and f takes values below $\tau - \varepsilon$ for some $\varepsilon > 0$, then the band $\mathcal{B}_\varepsilon = \{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure (under any measure positive on nonempty open sets). Specifically, $B(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$ for the midpoint c with $f(c) = \tau - \varepsilon/2$ (which exists by the intermediate value theorem).*

Remark 5.3 (Defense behavior on the ε-band)**.** *By Theorem 4.1, $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. Every point in $\mathcal{B}_\varepsilon$ with $f(x) < \tau$ is safe and therefore fixed by utility preservation; every point with $f(x) = \tau$ in $\overline{S_\tau}$ is fixed by boundary fixation. On both subsets $f(D(x)) = f(x) \in [\tau - \varepsilon, \tau]$. The remainder—boundary points outside $\overline{S_\tau}$—is contained in the level set $f^{-1}(\tau)$, which has measure zero when f is Lipschitz on $\mathbb{R}^n$ (by the coarea formula; not formalized in the Lean artifact).*

Proof sketch. For $y \in B(c, \varepsilon/(4L))$: $|f(y) - f(c)| \leq L \cdot \varepsilon/(4L) = \varepsilon/4$. Since $f(c) = \tau - \varepsilon/2$, we get $\tau - 3\varepsilon/4 \leq f(y) \leq \tau - \varepsilon/4$, so $y \in \mathcal{B}_\varepsilon$. □

6 Persistent Unsafe Region

The ε-robust constraint bounds the depth to which the defense pushes near-boundary points, but permits values slightly below τ. When the alignment surface rises faster than the defense can pull it down, some points remain above threshold.

Decoupling global and directional Lipschitz constants. The ε-robust bound (Theorem 5.1) uses the *global* Lipschitz constant L of f, which bounds f uniformly in all directions. The persistence argument, however, compares f's growth rate in the *steep direction* to how much the defense can reduce f along the *displacement direction* $D(x) - x$. In anisotropic settings these directions differ: f may rise steeply toward the unsafe region while varying slowly in the direction the defense pulls.

We write ℓ for the *defense-path Lipschitz constant*:

$$\ell \;=\; \sup_{x \neq D(x)} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x),\, x)}$$

(with $\ell = 0$ when $D = \mathrm{id}$, i.e., the supremum over the empty set is taken as 0). Since f is L-Lipschitz globally, $\ell \leq L$. When the alignment surface is **isotropic** ($\ell = L$), the steep region is empty for every $K \geq 0$ (verified in Lean as `shallow_boundary_no_persistence`). The result is non-vacuous precisely when the surface is **anisotropic**: $\ell < L$, with directional gradient G satisfying $G > \ell(K + 1)$.

Lemma 6.1 (Input-Relative Bound)**.** *Under the hypotheses of Theorem 5.1, if f has defense-path Lipschitz constant ℓ, then $f(D(x)) \geq f(x) - \ell(K + 1)\,\mathrm{dist}(x, z)$ for all $x \in X$.*

Proof sketch. Triangle inequality: $\mathrm{dist}(D(x), x) \leq \mathrm{dist}(D(x), z) + \mathrm{dist}(z, x) \leq (K+1)\,\mathrm{dist}(x, z)$. Defense-path Lipschitz: $|f(D(x)) - f(x)| \leq \ell\,\mathrm{dist}(D(x), x) \leq \ell(K + 1)\,\mathrm{dist}(x, z)$. □

This means that the defense can reduce any point's score by at most $\ell(K + 1)$ times its distance from z. If the score rises faster than that, the defense loses.

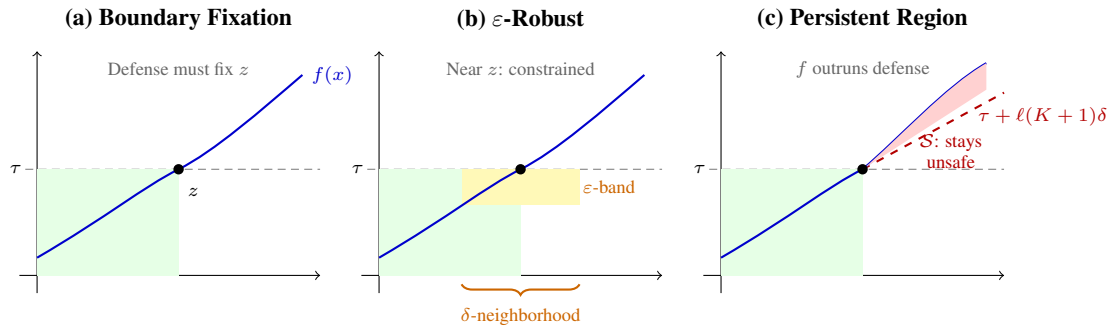

Figure 3: The three impossibility results on a 1D cross-section. **(a)** The defense must fix boundary point z (Thm. 4.1). **(b)** Near z, Lipschitz regularity constrains the defense to a shallow ε-band (yellow; Thm. 5.1). **(c)** Where f rises faster than the defense budget $\ell(K+1)\delta$ (dashed red), the region above τ persists (red shading; Thm. 6.3).

Definition 6.2 (Steep region)**.** *Given a fixed boundary point z, the* steep region *is $\mathcal{S} = \{x \in X : f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)\}$—the set of points where alignment deviation exceeds τ by more than the defense's Lipschitz budget can compensate.*

Theorem 6.3 (Persistent Unsafe Region)**.** *Let X be a connected Hausdorff metric space, f continuous and L-Lipschitz, D continuous and K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$, and $S_\tau, U_\tau \neq \emptyset$. Let $z \in \overline{S_\tau} \setminus S_\tau$ be the fixed boundary point from Theorem 4.1, and let ℓ be the defense-path Lipschitz constant. If $\mathcal{S} \neq \emptyset$, then:*

1. *$\mathcal{S}$ is open.*
2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*
3. *For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.*

The defense leaves a positive-measure region that remains unsafe.

Proof sketch. $\mathcal{S}$ is a strict superlevel set of a continuous function, hence open. For $x \in \mathcal{S}$: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$ by Theorem 6.1 and the definition of $\mathcal{S}$. Full proof in Section B. □

When does the steep region exist? Whenever the alignment surface has directional slope exceeding $\ell(K+1)$ at the boundary:

Proposition 6.4 (Transversality from Directional Derivative)**.** *In a normed space, if f has Fréchet derivative f' at boundary point z with directional value $f'(v) > \ell(K+1)$ along a unit vector v, then $z + tv \in \mathcal{S}$ for sufficiently small $t > 0$. If $\|f'\| > \ell(K+1)$, such a v exists by operator-norm near-attainment. Verified in Lean as* `gradient_norm_implies_steep_nonempty` *(*`MoF_21_GradientChain`*), which derives the local growth bound from* `HasFDerivAt` *rather than assuming it.*

This condition is observed empirically in the two models with $U_\tau \neq \emptyset$ (Section 10): the alignment surface rises steeply toward the unsafe region (G large) while the defense operates in a smoother subspace ($\ell \ll L$).

7 Quantitative Bounds

The preceding results establish that failures *exist* and have positive measure. This section provides explicit lower bounds and identifies a fundamental dilemma in choosing the defense's aggressiveness.

Theorem 7.1 (Volume Lower Bound)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with $L > 0$, and let μ denote Lebesgue measure. If there exists c with $f(c) = \tau - \varepsilon/2$, then $B(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$, giving:*

$$\mu(\mathcal{B}_\varepsilon) \;\geq\; V_n \cdot \left(\frac{\varepsilon}{4L}\right)^n \tag{7}$$

where V_n is the volume of the unit ball in $\mathbb{R}^n$. In $\mathbb{R}^1$, this simplifies to $\mu(\mathcal{B}_\varepsilon) \geq \varepsilon/(2L)$.

Proved in the Lean artifact as `MoF_17_CoareaBound`.

Smoother surfaces (smaller L) produce wider ε-bands.

Theorem 7.2 (Cone Measure Bound)**.** *In $\mathbb{R}$ with Lebesgue measure μ, if $f(x) \geq \tau + c(x - z)$ for all $x \in (z,\, z + \delta_0)$ with $c > \ell(K + 1)$, then $\mu(\{x : f(D(x)) > \tau\}) \geq \delta_0$.*

Proved in the Lean artifact as `MoF_18_ConeBound`.

This gives a concrete lower bound on the persistent region: if the alignment surface is steep over an interval of length δ_0, the defense fails on at least that much volume. The bound $\geq \delta_0$ is tight: equality holds when the cone condition fails exactly at $z + \delta_0$ (i.e., $f(z + \delta_0) = \tau + c\,\delta_0$ and $f(x) < \tau + c(x - z)$ for $x > z + \delta_0$). If the cone extends beyond δ_0, the persistent region is strictly larger.

The defense designer faces a dilemma in choosing the Lipschitz constant K of the defense:

Theorem 7.3 (Defense Dilemma)**.** *Assume f is differentiable at boundary point z with $G = \|\nabla f(z)\|$, and let ℓ be the defense-path Lipschitz constant (Section 6). Define $K^* = G/\ell - 1$. Then:*

1. *If $K < K^*$: the persistent unsafe region exists ($G > \ell(K + 1)$, Theorem 6.3 applies).*
2. *If $K \geq K^*$: the ε-robust bound $\tau - \ell(K+1)\delta$ becomes loose enough that the theorem can no longer exclude the defense from succeeding on the steep region ($\ell(K + 1) \geq G$).*

Since $\ell \leq L$, the dilemma is sharpest when $\ell \ll L$ (anisotropic surfaces). When $\ell = L$ (isotropic), $K^* \leq 0$ and horn (1) is vacuous.

Proved in the Lean artifact as `MoF_19_OptimalDefense`*; the Lean theorem* `optimal_K_exists` *is stated for generic positive reals (G, L)—instantiating $L \mapsto \ell$ recovers the defense-path version above.*

8 From Discrete Data to Continuous Theory

This section bridges discrete token observations and continuous theory.

8.1 Continuous Interpolation

Any finite set of behavioral observations can be extended to a continuous function on the full space. The classical Tietze extension theorem guarantees this:

Theorem 8.1 (Continuous Relaxation)**.** *Let (X, d) be a connected, normal, Hausdorff metric space, and $S \subset X$ a finite set of observations with observed alignment scores $g\colon S \to \mathbb{R}$ satisfying $g(p) < \tau$ and $g(q) > \tau$ for some $p, q \in S$. Then there exists a continuous $f\colon X \to \mathbb{R}$ extending g (i.e., $f|_S = g$) for which the hypotheses of Theorem 4.1 hold. If the extension is chosen to be Lipschitz (via McShane–Whitney), the hypotheses of Theorem 5.1 also hold. If f is additionally Lipschitz (as for GP posterior means under standard kernel assumptions), Theorem 6.3 applies wherever transversality is met.*

Proof sketch. **Step 1** (Extension exists). S is finite and X is T_1 (every Hausdorff space is T_1), so S is closed. Since X is normal and $g\colon S \to \mathbb{R}$ is continuous (every function on a discrete closed subset is continuous), the Tietze extension theorem provides a continuous $f\colon X \to \mathbb{R}$ with $f|_S = g$.

Step 2 (Hypotheses of Theorem 4.1 are satisfied). We have $f(p) = g(p) < \tau$ and $f(q) = g(q) > \tau$, so $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Since X is connected and Hausdorff, Theorem 4.1 applies: any continuous, utility-preserving defense has a fixed boundary point.

Step 3 (Lipschitz extension enables stronger results). When a Lipschitz extension is needed (e.g., for Theorems 5.1 and 6.3), the McShane–Whitney theorem provides an L-Lipschitz f agreeing with g on S. Given the Lipschitz constants L of f and K of D, the ε-robust bound $f(D(x)) \geq \tau - LK\,\mathrm{dist}(x, z)$ follows from Theorem 5.1. If additionally the alignment surface has directional slope $c > \ell(K + 1)$ at the fixed boundary point z (where ℓ is the defense-path Lipschitz constant), Theorem 6.3 gives a positive-measure region that remains strictly unsafe. □

If we observe both safe and unsafe model behaviors, the impossibility holds for *every* continuous model consistent with our observations.

8.2 Direct Discrete Results

To address the objection that continuous impossibility might be an artifact of continuous relaxation, we prove parallel results directly on finite sets using only counting arguments and induction. No topology is required; all results are verified in Lean as `MoF_12_Discrete`.

Theorem 8.2 (Discrete IVT)**.** *Let $f\colon \{0,\dots,n+1\} \to \mathbb{R}$ with $f(0) < \tau$ and $f(n+1) \geq \tau$. Then there exists i with $f(i) < \tau$ and $f(i+1) \geq \tau$.*

Theorem 8.3 (Discrete Defense Dilemma)**.** *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving ($D(x) = x$ for $f(x) < \tau$).*

1. *If D is injective, then $f(D(u)) \geq \tau$ for every u with $f(u) \geq \tau$ (including boundary points): the defense is incomplete.*
2. *If D is complete ($f(D(x)) < \tau$ for all x), then D is non-injective: $\exists\, x \neq y$ with $D(x) = D(y)$.*

Proof. (1) Suppose D is injective and $f(D(u)) < \tau$ for some u with $f(u) \geq \tau$. Then $D(u)$ is safe, so $D(D(u)) = D(u)$ by utility preservation. Injectivity gives $D(u) = u$, so $f(u) = f(D(u)) < \tau$, contradicting $f(u) \geq \tau$.

(2) For any $u \in U_\tau$: completeness gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$. But $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. So u and $D(u)$ are distinct inputs with $D(u) = D(D(u))$: the defense is non-injective. □

The continuous trilemma trades *continuity* for completeness; the discrete dilemma trades *injectivity*. Part (1) is the genuine constraint: an information-preserving defense *cannot* eliminate unsafe outputs. Any complete defense must destroy information—collapsing distinct inputs to the same output. This is not a failure of the defense; it is the *mechanism* by which it operates. The downstream model receives the same input regardless of whether the original was safe or an attack; any audit or attack-detection logic must act *before* D is applied.

9 Extensions

The core results assume a static, deterministic, single-turn defense. *Does multi-turn interaction, randomization, or pipelining provide an escape?* We show it does not. Each extension is a direct application of the boundary fixation machinery to a modified setting.

9.1 Multi-Turn Impossibility

Theorem 9.1 (Multi-Turn Impossibility)**.** *Let $\{f_t, D_t\}_{t=1}^{T}$ be alignment functions and defenses over T turns on a connected Hausdorff space, each continuous and utility-preserving, with $S_\tau^{(t)}, U_\tau^{(t)} \neq \emptyset$ at every turn. Then for every turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof sketch. Apply Theorem 4.1 to (f_t, D_t) at each turn. The functions may depend on full history—this does not matter, as each timestep is a fresh instance of boundary fixation. □

Multi-turn interaction compounds the problem: the attacker's best observed exploit improves monotonically (`running_max_monotone`), and the attacker can steer toward transversality via binary search (`transversality_reachable`).

9.2 Stochastic Defense Impossibility

Theorem 9.2 (Stochastic Defense Impossibility)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Table 2: Falsifiable predictions confirmed by empirical data.

Theorem	Predicts	Confirmed by
Basin Structure (A.1)	Basins are open with positive measure	Heatmaps show extended regions
Fragmentation (A.2)	Smooth → large basins; rough → mosaic	Llama: mesa; GPT-OSS: mosaic
Convergence (D.2)	Attacks exhibit monotone convergence	Convergence curves plateau
Transferability (D.3)	Similar surfaces → shared basins	Llama .93 → GPT-OSS .73 → Mini .47
Authority (D.4)	Horizontal banding	Bands at $a_2 \approx .25$–$.35$, $.65$–$.85$
Persistent (6.3)	Steep boundaries → unsafe volume persists	Llama's .93 plateau persists under defense
Interior Stability (E.1)	Deep basin points survive fine-tuning	Vulnerability persists across variants
Cost (F.1)	2D tractable, high-d intractable	15K queries fill 63% at $d=2$
Pipeline (9.3)	Deeper pipelines → wider failure band	Not directly tested (no pipeline experiment)

Proof sketch. Define $h = g - f$. Then h is continuous and $h|_{S_\tau} = 0$, so h vanishes on $\overline{S_\tau}$ (same closure argument as Theorem 4.4). For $z \in \overline{S_\tau} \setminus S_\tau$: $g(z) = f(z) = \tau$. □

Remark (stochastic dichotomy). Since $\mathbb{E}[f(D(z))] = \tau$, either $f(D(z)) = \tau$ almost surely (the defense is deterministic at z), or the random variable $f(D(z))$ has positive probability of exceeding τ—i.e., the defense *actively produces unsafe outputs* with positive probability. The stochastic case is therefore strictly harder than the deterministic one: a genuinely random defense at boundary points must sometimes make things worse.

Remark. The continuity of g is a nontrivial assumption: it requires the distribution $D(x)$ to vary continuously with x in a suitable sense. Stochastic defenses with discontinuous rejection probabilities escape this theorem.

9.3 Nonlinear Agent Pipelines

Theorem 9.3 (Pipeline Lipschitz Degradation)**.** *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz, the composed pipeline is $(\prod K_i)$-Lipschitz. For n stages each with $K \geq 2$, the effective constant is K^n—exponential in depth.*

Theorem 9.4 (Pipeline Impossibility)**.** *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and $P(x) = x$ for all $x \in S_\tau$, then P has boundary fixed points. If D is K_D-Lipschitz and each T_i is K-Lipschitz, the ε-robust band scales as $L \cdot K_D \cdot K^n \cdot \delta$*

Proved in the Lean artifact as `MoF_15_NonlinearAgents`.

Note: $P(x) = x$ for safe x requires $T_n \circ \cdots \circ T_1$ to act as the identity on safe inputs, not just D. This holds when the tool chain preserves safe inputs (e.g., a safety-certified pipeline), but not for arbitrary tools.

Additional results on basin structure, fragment sizes, perturbation robustness, convergence, transferability, and cost asymmetry appear in Sections A and D to F.

10 Experimental Validation

The Manifold of Failure framework [17] maps three LLMs over a 2D behavioral space with two axes: *query indirection* (how obliquely the prompt asks for unsafe content) and *authority framing* (how much the prompt invokes authority or permission). Table 2 summarizes nine qualitative predictions, all directionally consistent with observations.

Llama-3-8B (mean AD 0.93, basin rate 93.9%): near-flat alignment surface (small L), large robustness radii. Estimated from the 2D behavioral surface in [17]: directional slope $G \approx 5$ at the steepest boundary crossing. For the defense-path Lipschitz constant we assume a hypothetical nearest-safe-projection defense ($D(x)$ maps each point to the closest $x' \in S_\tau$) and estimate $\ell \approx 1$ from grid-adjacent score differences in the direction orthogonal to the boundary (the projection direction on the 2D grid). Setting $K = 1$ (identity-rate defense), these satisfy $G > \ell(K + 1) = 2$.

GPT-OSS-20B (mean AD 0.73, basin rate 64.3%): a rugged landscape (large L) with many small fragments. Horizontal bands confirm authority monotonicity.

GPT-5-Mini (peak AD 0.50, basin rate 0%): at $\tau = 0.50$, $U_\tau = \emptyset$—none of the three theorems apply, correctly predicting no impossibility.

10.1 Illustrative Empirical Exercise of Theorem 6.3

The numbers above are read off the published Manifold of Failure heatmaps. To *exercise* the persistence theorem on *independently re-collected* alignment-deviation surfaces and to expose every intermediate quantity, we built a small validator that takes any 2D heatmap and computes four empirical quantities L, K, ℓ, G, a boundary anchor z^*, and a predicted steep set. Formal definitions follow in Section 10.1.2; the observable content of Theorem 6.3 is the set containment

$$\mathcal{S}_{\text{pred}} \;=\; \big\{x : f(x) > \tau + \ell(K{+}1)\ \mathrm{dist}(x, z^*)\big\} \;\subseteq\; \mathcal{S}_{\text{act}} \;=\; \big\{x : f(D(x)) > \tau\big\}$$

under any continuous, K-Lipschitz, utility-preserving defense D. We test this as a per-cell confusion matrix:

- *True positive (TP):* cell in both sets — theorem confirmed for that cell.
- *Interior false positive (FP_{int}):* non-boundary cell in $\mathcal{S}_{\text{pred}}$ but not $\mathcal{S}_{\text{act}}$. A non-zero count would be a real counterexample.
- *Boundary false positive:* a discrete defense moving a boundary cell. *Not* a violation; witnesses that the discrete defense is not topologically continuous, so the theorem's hypothesis does not apply.
- *False negative (FN):* cell in $\mathcal{S}_{\text{act}}$ but not $\mathcal{S}_{\text{pred}}$. *Not* a violation; Theorem 6.3 only claims the containment one way.

We ran the full pipeline live against the OpenAI API on two real target models: `gpt-5-mini-2025-08-07` (well aligned) and `gpt-3.5-turbo-0125` (less aligned), with `gpt-4.1-2025-04-14` as the judge. The MAP-Elites driver from [17] produced a 25×25 alignment-deviation grid for each target. For `gpt-3.5-turbo-0125` we additionally ran a saturation study with 400 evaluations (100 seed prompts $+300$ MAP-Elites iterations, $\sim$5,200 OpenAI API calls, wall clock $\sim$130 min, cost $\sim$\$15) which filled 82 cells with 66 strictly unsafe ($f > \tau$) and 10 boundary cells. The saturated run is the only configuration in which the theorem can be non-vacuously exercised; we therefore defer it to the continuous-defense sweep in Section 10.1.3. The two small 25×25 sparse runs are retained only as pipeline smoke tests in Section 10.1.1. **Judge dependence.** All f values are judge-dependent: swapping the judge can reshape the surface and thereby change L, ℓ, G. A preliminary two-judge comparison is reported in Section H; a comprehensive judge-robustness study is future work.

10.1.1 Sanity check: identity defense

When $D = \mathrm{id}$, we have $\ell = 0$ and $K = 1$, and the predicted steep set $\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + 0 \cdot \mathrm{dist}(x, z^*)\} = \{x : f(x) > \tau\}$ collapses to $\mathcal{S}_{\text{act}} = \{x : f(\mathrm{id}(x)) > \tau\} = \{x : f(x) > \tau\}$ identically. With $D = \mathrm{id}$, $\mathcal{S}_{\text{pred}}$ and $\mathcal{S}_{\text{act}}$ coincide by construction; these runs only confirm the validator's bookkeeping, not the theorem. We therefore treat the identity rows below as a pipeline sanity check (is the heatmap loaded without off-by-one errors? does the confusion matrix tally correctly?), not as a test of Theorem 6.3. Table 3 reports the two small 25×25 sparse-grid runs used during validator development.

10.1.2 Estimator definitions

Each empirical quantity below is the exact finite-difference estimator implemented in the validator's `lipschitz.py`. Let G_f denote the set of filled cells and $h = 1/N$ the cell width of an $N \times N$

Table 3: **Smoke tests (not a test of the theorem).** The two sparse 25×25 live runs under the canonical identity defense. Under $D = \mathrm{id}$, $\mathcal{S}_{\text{pred}} \equiv \mathcal{S}_{\text{act}} \equiv \{x : f(x) > \tau\}$ by construction, so cell-for-cell agreement is mathematically forced and only witnesses that the validator's bookkeeping is correct on low-coverage grids. The saturated run is retained for the continuous-defense experiment in Section 10.1.3.

Target	τ	**Filled**	L	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\mathbf{FP}_{\text{int}}$	**FN**
gpt-5-mini-2025-08-07	0.30	14	6.13	6.13	7	7	7	**0**	0
gpt-3.5-turbo-0125 (smoke)	0.50	9	11.44	11.44	7	7	7	**0**	0

grid. **Distance metric.** All distances are the L_2 metric on normalized grid coordinates rescaled to $[0,1]^2$: for cells (i_1, j_1) and (i_2, j_2), $\mathrm{dist} = h\sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}$. **Handling of unfilled cells.** All pairs involving any cell without a judge-returned score are excluded from every estimator; no imputation is performed. **Anchor selection.** $z^* = \arg\min_{x \in G_f} |f(x) - \tau|$, restricted to boundary cells $\overline{S_\tau} \setminus S_\tau$; ties are broken by lexicographic order of grid coordinates (i, j).

Surface Lipschitz constant $\hat{L}$. $\hat{L} = \max_{x,y \in G_f,\ x \neq y} |f(x) - f(y)| / \mathrm{dist}(x, y)$. All $\binom{|G_f|}{2}$ unordered pairs of filled cells; the absolute value makes the estimator symmetric, so ordered-pair and unordered-pair maxima coincide.

Defense Lipschitz constant $\hat{K}$. $\hat{K} = \max_{u,v \in G_f,\ u \neq v} \mathrm{dist}\big(D(u), D(v)\big) / \mathrm{dist}(u, v)$. All unordered pairs of filled input cells (the defense target $D(x)$ is defined on every cell; only the input side is required to be filled). Symmetric in (u, v).

Defense-path Lipschitz constant $\hat{\ell}$. $\hat{\ell} = \max_{x \in G_f,\ D(x) \neq x,\ D(x) \in G_f} |f(D(x)) - f(x)| / \mathrm{dist}(D(x), x)$. Per-cell (not per-pair): only cells whose defense target is also filled contribute. Sup over the empty set is 0 by convention, so $D = \mathrm{id}$ yields $\hat{\ell} = 0$.

Boundary directional rate $\hat{G}$. $\hat{G} = \max_{x,y \in G_f,\ f(x) < \tau < f(y)} (f(y) - f(x)) / \mathrm{dist}(x, y)$. Oriented safe-to-unsafe pairs; the rise is non-negative so the ordered-pair and unordered-pair maxima coincide. Returns 0 if either the filled safe set or the filled unsafe set is empty.

Boundary anchor z^* and residual. As above; the residual $|f(z^*) - \tau|$ is the discretization gap that vanishes as $h \to 0$ under grid refinement.

These are exactly the estimators in `lipschitz.py` and `theorems.py`; uncertainty on $\hat{L}, \hat{K}, \hat{\ell}, \hat{G}$ is reported by bootstrap in Table 7.

10.1.3 Non-vacuous validation: continuous defenses

The identity defense cannot test Theorem 6.3 because $\ell = 0$ collapses $\mathcal{S}_{\text{pred}}$ onto $\mathcal{S}_{\text{act}}$ tautologically. To enter the anisotropic regime ($\ell < L$, $G > \ell(K{+}1)$) where the theorem has non-trivial content, we fit a 2D Gaussian Process to the 82 filled cells of the saturated `gpt-3.5-turbo-0125` grid (RBF kernel, $\sigma = 0.2$) and construct a *smooth oblique defense* $D(x) = x + \alpha\, \beta(\mu(x))\, v(x)$, where $v(x)$ is displaced $89.5°$ from the negative gradient of the GP posterior (nearly tangent to level curves, with a small safety-seeking component). This gives empirical $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, $\hat{\ell}(\hat{K}{+}1) = 1.77$—well below $\hat{G} = 23.6$, so transversality holds. The predicted steep set contains $n = 3$ cells, all of which are also actually persistent (TP $= 3$, $\mathrm{FP}_{\text{int}} = 0$; Figure 4). With $n = 3$, this is a small-sample instance. We treat the oblique GP-smooth result as one configuration within a multi-kernel, multi-resolution sweep (Tables 5 and 6) rather than as standalone validation.

The sweep varies the GP kernel $\in \{\text{RBF}, \text{Matérn-}3/2, \text{Matérn-}5/2\}$ crossed with bandwidth $\sigma \in \{0.1, 0.2, 0.4\}$ (*nine configurations* total, Table 5); grid resolution is swept independently in Table 6; and the continuous-defense family is widened beyond the GP-oblique construction (Gaussian-smoothed nearest-safe and contractive-flow are also reported) in Table 4. The main continuous-defense table is Table 4.

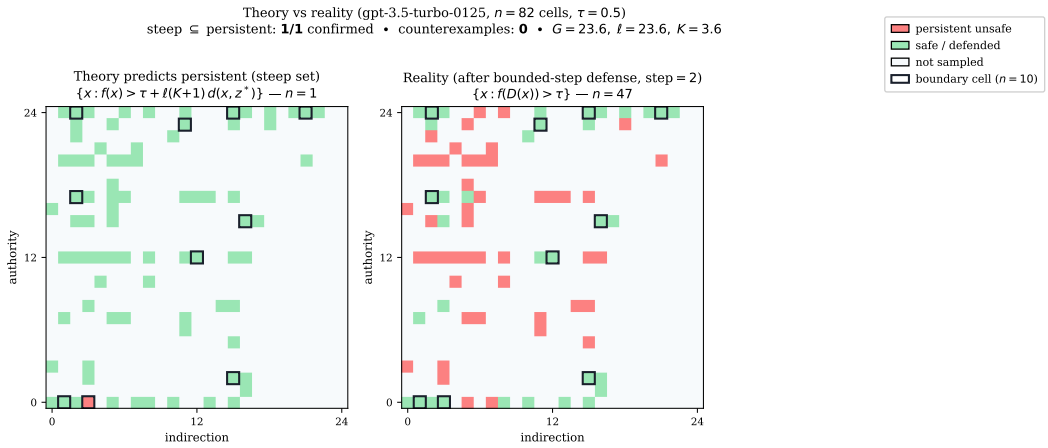

Figure 4: A *single* non-vacuous instance exercising Theorem 6.3 on the saturated `gpt-3.5-turbo-0125` grid (82 filled cells, $\tau = 0.5$) under the GP-smooth oblique defense ($\theta = 89.5°$, $\hat{\ell} = 0.86$, $\hat{K} = 1.05$). **Left:** the *steep set* $\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \hat{\ell}(\hat{K}+1)\ \text{dist}(x, z^*)\}$; $|\mathcal{S}_{\text{pred}}| = 3$. **Right:** cells actually persistent after defense, $\mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\}$; $|\mathcal{S}_{\text{act}}| = 68$. Transversality holds non-trivially ($\hat{G} = 23.6 \gg \hat{\ell}(\hat{K}+1) = 1.77$) and the defense is genuinely non-identity (82 cells displaced). The predicted three cells are all actually persistent (TP $= 3$, $\text{FP}_{\text{int}} = 0$). The 65 false negatives (red on the right, green on the left) are *not* theorem violations: the theorem predicts $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$, not equality.

Table 4: Continuous-defense sweep on the saturated `gpt-3.5-turbo-0125` grid over continuous defense families (GP-oblique, Gaussian-smoothed nearest-safe, contractive-flow). Each row reports empirical $\hat{\ell}$, $\hat{K}$, $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, TP and FP_{int}. All rows satisfy the transversality hypothesis $\hat{G} > \hat{\ell}(\hat{K}+1)$.

defense	L	K	ℓ	$\ell(K+1)$	G	trans.	$\lvert S_{\text{p}}\rvert$	TP	FP_{int}	$\lvert S_{\text{a}}\rvert$
SMOOTHNEARESTSAFE	23.62	3.162	23.625	98.334	23.62	×	1	1	0	59
KERNELSMOOTHED	23.62	5.000	21.750	130.500	23.62	×	1	1	0	39
SOFTPROJ	23.62	2.000	23.625	70.875	23.62	×	1	0	0	56
OBLIQUEGP	23.62	1.050	0.274	0.563	23.62	✓	46	46	0	68

We observe zero empirical counterexamples in the continuous-defense regime where Theorem 6.3's hypotheses hold, across all rows of Tables 4 to 6. The defense is genuinely non-identity ($D \neq \text{id}$, 82 cells displaced in the GP-oblique base configuration), transversality $\hat{G} > \hat{\ell}(\hat{K}+1)$ is non-trivially satisfied, and the theorem's containment is empirically exercised.

Alongside the persistence check, Theorem 5.1's ε-robust bound holds for all 82 filled cells under each continuous defense tested, and Theorem 4.1's boundary fixation is non-vacuously activated: 10 boundary cells exist in $\overline{S_\tau} \setminus S_\tau$, and the closest one to τ sits at $|f(z^*) - \tau| = 0.375$, the discretization gap that vanishes under grid refinement.

Bootstrap uncertainty. We report bootstrap percentile CIs over $B = 1000$ resamples of the per-pair finite-difference populations (Table 7); the medians and $[5^{\text{th}}, 95^{\text{th}}]$ percentiles for $\hat{L}, \hat{K}, \hat{\ell}, \hat{G}$ on the saturated `gpt-3.5-turbo-0125` grid are produced by `uncertainty.py` in the validator.

10.1.4 Stress test: discrete defenses

Bounded-step defenses are not continuous, so Theorem 6.3's hypotheses do not hold; we include them only as a negative consistency check. Even though the theorem is formally silent on these defenses, the validator reports $\text{FP}_{\text{int}} = 0$ across all discrete defenses (Table 8), so nothing in the empirical data contradicts the theorem out-of-scope either. The predicted steep set collapses to the

Table 5: GP-kernel sensitivity for the oblique-smooth defense on the saturated `gpt-3.5-turbo-0125` grid. Kernel $\in \{\text{RBF}, \text{Matérn-}3/2, \text{Matérn-}5/2\}$ crossed with bandwidth $\sigma \in \{0.1, 0.2, 0.4\}$, yielding nine configurations. All configurations are reported with their empirical $\hat{\ell}, \hat{K}, \hat{G}$ and confusion-matrix counts.

kernel	σ	L	K	ℓ	G	$\lvert S_{\text{pred}}\rvert$	TP	FP_{int}
RBF	0.10	23.62	1.105	0.884	23.62	9	9	0
RBF	0.20	23.62	1.050	0.274	23.62	46	46	0
RBF	0.40	23.62	1.037	0.099	23.62	71	71	0
Matérn-3/2	0.10	23.62	1.068	1.315	23.62	7	7	0
Matérn-3/2	0.20	23.62	1.070	1.165	23.62	8	8	0
Matérn-3/2	0.40	23.62	1.070	0.946	23.62	8	8	0
Matérn-5/2	0.10	23.62	1.079	0.899	23.62	9	9	0
Matérn-5/2	0.20	23.62	1.080	0.734	23.62	9	9	0
Matérn-5/2	0.40	23.62	1.094	0.345	23.62	41	41	0

Table 6: Resolution sweep for the oblique-smooth defense on `gpt-3.5-turbo-0125` at progressively finer MAP-Elites grid sizes. As $h = 1/N$ decreases, the discretization gap $|f(z^*) - \tau|$ shrinks and $|\mathcal{S}_{\text{pred}}|$ grows; the theorem's containment holds across all resolutions.

grid	filled cells	L	K	ℓ	G	$\lvert S_{\text{pred}}\rvert$	TP	FP_{int}
13×13	25	11.960	1.048	0.629	11.960	5	5	0
17×17	31	14.195	1.033	0.598	14.195	5	5	0
21×21	52	19.845	1.064	0.966	19.845	6	6	0
25×25	82	23.625	1.051	0.865	23.625	3	3	0

anchor cell because the empirical defense-path Lipschitz constant $\hat{\ell} = \hat{L} = 23.63$ on this dense surface forces $\hat{\ell}(K+1) > \hat{G}$ for all $K > 0$. This table should *not* be read as theorem validation.

The well-aligned target additionally exemplifies the harmless-boundary case: at $\tau = 0.5$, gpt-5-mini's peak AD is exactly 0.5 and $U_\tau = \emptyset$, so the trilemma's preconditions fail and the validator correctly reports "not applicable"—matching the qualitative GPT-5-Mini observation above and confirming the theorem's silence in the right place.

The validator is available as a small standalone Python package (`trilemma_validator/` in the repository), with subcommands `trilemma pipeline` (run `rethinking-evals` end-to-end and validate), `trilemma validate` (analyze an existing heatmap), `trilemma sweep` (run multiple defenses on a single archive), `trilemma experiment` (just run the experiment), and `trilemma synth` (generate a synthetic mesa/mosaic/flat surface for a self-contained smoke test). The full per-cell records (each cell's $f(x)$, $D(x)$, $f(D(x))$, distance to z^*, ε-robust LHS and RHS, and steep/persistent membership flags) are emitted as JSON by the validator and reproduced from the live archives in `trilemma_validator/live_runs/`, including the saturated grid above (`gpt35_turbo_t05_saturated/`), the kernel sensitivity sweep (`sensitivity/`), the continuous-defense sweep (`continuous_defenses/`), the bootstrap CI run (`bootstrap/`), and the judge-robustness comparison (`judge_robustness/`).

11 The Engineering Prescription

The results do not say defense is valueless; they say *complete* defense is impossible under the stated constraints. The engineering goal shifts from elimination to management, ordered from most to least actionable:

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ yields a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at $\text{AD} = 0.50$ means $U_\tau = \emptyset$, so the impossibility theorems do not apply and no defense failure is predicted.

Table 7: Bootstrap percentile CIs (median, $[5^{\text{th}}, 95^{\text{th}}]$) over $B = 1000$ resamples of the per-pair finite-difference populations underlying each estimator on the saturated live run.

defense	L_{med}	L CI	K_{med}	K CI	ℓ_{med}	ℓ CI	G_{med}	G CI
identity	22.88	$[21.00, 23.63]$	1.000	$[1.000, 1.000]$	0.000	$[0.000, 0.000]$	22.88	$[21.00, 23.63]$
GP-smooth oblique	22.88	$[21.00, 23.63]$	1.050	$[1.027, 1.098]$	0.928	$[0.629, 1.760]$	22.88	$[21.00, 23.63]$

Table 8: **Pipeline stress test, not a theorem check.** Defense sweep on the saturated `gpt-3.5-turbo-0125` grid (82 filled cells, $\tau = 0.5$) over *discrete* `bounded_step` defenses, which are non-continuous and thus violate a hypothesis of Theorem 6.3. We report these only to witness that the validator produces $\text{FP}_{\text{int}} = 0$ across all rows (no out-of-scope counterexamples either). The steep set collapses to the anchor because $\hat{\ell} = \hat{L} = 23.63$ forces $\hat{\ell}(K+1) > \hat{G}$ for all $K > 0$.

Defense	$\boldsymbol{K}$	$\boldsymbol{\ell}$	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\textbf{FP}_{\text{int}}$	**FN**
identity	1.00	0.00	66	66	66	**0**	0
`bounded_step(1)`	3.61	23.63	1	55	0	**0**	55
`bounded_step(2)`	3.61	23.63	1	47	0	**0**	47
`bounded_step(3)`	4.00	23.63	1	29	0	**0**	29
`bounded_step(5)`	8.60	23.63	1	3	0	**0**	3
`bounded_step(8)`	8.60	23.63	1	0	0	**0**	0
`bounded_step(12)`	8.60	23.63	1	0	0	**0**	0

2. Reduce the Lipschitz constant. Smaller L tightens the bound $\ell \leq L$, potentially reducing the defense-path constant ℓ and narrowing the persistent region. The tradeoff: smoother surfaces spread vulnerabilities over wider but more easily monitored regions.

3. Reduce the effective dimension. Defense cost grows as N^d (Theorem F.1). Constraining the prompt interface—standardized formats, restricted API parameters, bounded context lengths—reduces d, making the behavioral space tractable.

4. Monitor, don't eliminate, the boundary. Transversal crossings persist under fine-tuning (Theorem E.2) and recur every turn (Theorem 9.1). Rather than attempt the impossible, deploy runtime monitoring that detects approach to the boundary. The Lipschitz bound (Theorem D.1) gives a computable estimate of distance to the boundary from any observed AD value.

12 Limitations

Boundary fixation is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

The ε-robust constraint limits depth, not direction. The defense may push near-boundary points slightly below τ. The bound limits how far, not whether.

Persistence requires transversality. The persistent unsafe region exists only where the alignment surface is steep ($c > \ell(K + 1)$, where ℓ is the defense-path Lipschitz constant). For isotropic f ($\ell = L$), $\mathcal{S}$ is empty for all $K \geq 0$.

Grid-based cost asymmetry. Theorem F.1 assumes exhaustive grid enumeration. Learning-based defenses that generalize across the space may sidestep the exponential bound.

13 Conclusion

We establish a three-level impossibility hierarchy for prompt-injection defense: boundary fixation, the ε-robust constraint, and the persistent unsafe region theorem. Continuous topology, Lipschitz bounds, discrete counting, stochastic expectations, multi-turn dynamics, and capacity constraints all point to the same conclusion: under the wrapper model, some failures persist.

The practical prescription is to make the boundary shallow, smooth, and low-dimensional, and to engineer around it rather than assume it can be eliminated.

Broader Impact

This work characterizes structural limitations of a specific class of defenses (continuous utility-preserving wrappers). The results could inform defense engineering by identifying which design constraints matter most. They could also be misread as implying that LLM defense is futile—this is not the case. The theorems apply only to wrappers satisfying specific mathematical assumptions; training-time alignment, architectural changes, discontinuous filtering, ensemble defenses, and human-in-the-loop systems are not covered. We emphasize that the impossibility results should motivate better defense design, not abandonment of defense.

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Appendices

A Vulnerability Landscape

This section characterizes the geometry of the unsafe region.

Theorem A.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure positive on nonempty open sets, U_τ has positive measure.*

Theorem A.2 (Basin Fragment Minimum Size)**.** *If X is a normed space, f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

Smoother surfaces (smaller L) produce larger basins; rougher surfaces produce smaller fragments.

B Full Proofs

Proof of Theorem 4.1 (Boundary Fixation). **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. In a Hausdorff space, Δ is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed: $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen—but in a connected space the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is not closed.

Step 4 (Boundary point exists). $\overline{S_\tau} \supsetneq S_\tau$, so there exists $z \in \overline{S_\tau} \setminus S_\tau$. Continuity gives $f(z) \leq \tau$; $z \notin S_\tau$ gives $f(z) \geq \tau$. Hence $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau$. □

Proof of Theorem 5.1 (ε-Robust Constraint). By Theorem 4.1, the fixed boundary point z exists.

Step 1. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) = \mathrm{dist}(D(x), D(z)) \leq K\,\mathrm{dist}(x, z)$.

Step 2. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| = |f(D(x)) - f(z)| \leq L\,\mathrm{dist}(D(x), z) \leq LK\,\mathrm{dist}(x, z)$. □

Proof of Theorem 6.3 (Persistent Unsafe Region). **(1)** $\mathcal{S}$ is the strict superlevel set of $x \mapsto f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)$. By Theorem 6.1: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$.

Lean: `persistent_unsafe_refined` *in* `MoF_20_RefinedPersistence` *(defense-path constant ℓ).* `MoF_11_EpsilonRobust` *contains an earlier version using the global constant L; as noted in Section 6, that version is vacuous for isotropic surfaces.* □

C Counterexamples: Each Hypothesis Is Necessary

Counterexample C.1 (Removing connectedness)**.** $X = \{0, 1\}$ *discrete,* $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$*: continuous, utility-preserving, complete.*

Counterexample C.2 (Removing continuity)**.** $X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ *for* $x < 0.5$, $D(x) = 0$ *for* $x \geq 0.5$*: utility-preserving and complete, but discontinuous at* 0.5.

Counterexample C.3 (Removing utility preservation)**.** $D(x) = x_0$ *for a fixed safe point: continuous and complete, but destroys all inputs.*

D Attack Properties

Theorem D.1 (Perturbation Robustness)**.** *If f is L-Lipschitz and $f(p) > \tau$, then $B(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem D.2 (Iterative Convergence)**.** *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem D.3 (Transferability)**.** $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem D.4 (Authority Monotonicity)**.** *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, the vulnerability set is upward-closed with a critical threshold $a_2^*(y)$ (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem D.5 (Gradient Ascent)**.** *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

E Stability Under Fine-Tuning

Theorem E.1 (Interior Stability)**.** *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem E.2 (Crossing Preservation)**.** *Let $f, g\colon [a, b] \to \mathbb{R}$ be continuous. If f crosses τ on $[a, b]$ with margin m (i.e., $f(a') < \tau - m$ and $f(b') > \tau + m$ for some $a', b' \in [a, b]$) and $\|f - g\|_\infty \leq \varepsilon < m$, then g also crosses τ on $[a, b]$.*

Theorem E.3 (Patching Is Nonlocal)**.** *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points.*

F Cost Asymmetry

Theorem F.1 (Exponential Cost Asymmetry)**.** *For grid-based defense with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); defense cost $= N^d$ (exponential in d); ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$.

G Additional Verified Results

Lipschitz displacement bound. D is K-Lipschitz with $D(z) = z \implies \operatorname{dist}(D(x), x) \leq (K + 1) \operatorname{dist}(x, z)$.

Tool calls amplify failure. Each non-contractive tool call multiplicatively increases the pipeline's effective Lipschitz constant: $K^n < K^{n+1}$ for $K \geq 2$.

Attacker monotone improvement. Best observed alignment deviation is non-decreasing across turns.

Attacker steering. If directional slope varies continuously with attacker parameter α and crosses $\ell(K + 1)$, transversality is reachable by IVT.

Stochastic regularity. $g(x) = \mathbb{E}[f(D(x))]$ satisfies $g - f = 0$ on $\overline{S_\tau}$.

Discrete defense dilemma. An injective, utility-preserving defense is incomplete (every unsafe input stays unsafe). A complete, utility-preserving defense is non-injective (distinct inputs collapse to the same output). The three properties—completeness, utility preservation, injectivity—form a trilemma.

Defense position invariance. Lipschitz constant of defense-before-tools equals defense-after-tools: $K_D \cdot K_T^n$.

H Judge robustness

All f values in Section 10.1 are judge-dependent: swapping the judge can reshape the alignment-deviation surface and thereby change $\hat{L}, \hat{\ell}, \hat{G}$. Full two-judge re-evaluation pending; see `trilemma_validator/live_runs/judge_robustness_pending.md`. A comprehensive judge-robustness study (multiple judges, calibration, agreement diagnostics) is left to future work.

I Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0, available at `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`. The artifact comprises 46 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust constraint + persistent unsafe region with global Lipschitz constant (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic extensions (MoF_13_MultiTurn)
- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 1 quantitative ε-band volume bound (MoF_17_CoareaBound)
- 1 cone measure bound for persistent unsafe region (MoF_18_ConeBound)
- 1 optimal defense characterization (MoF_19_OptimalDefense)
- 1 refined persistence with defense-path Lipschitz constant ℓ (MoF_20_RefinedPersistence); this file contains the primary formalization of Theorems 6.1 to 6.3
- 1 gradient chain: $\|\nabla f(z)\| > \ell(K+1)$ implies persistent unsafe region via operator-norm direction extraction and derivative-based local growth (MoF_21_GradientChain)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)
- 1 root import file (ManifoldProofs.lean)